

On the instability mechanism of a slab microtearing mode

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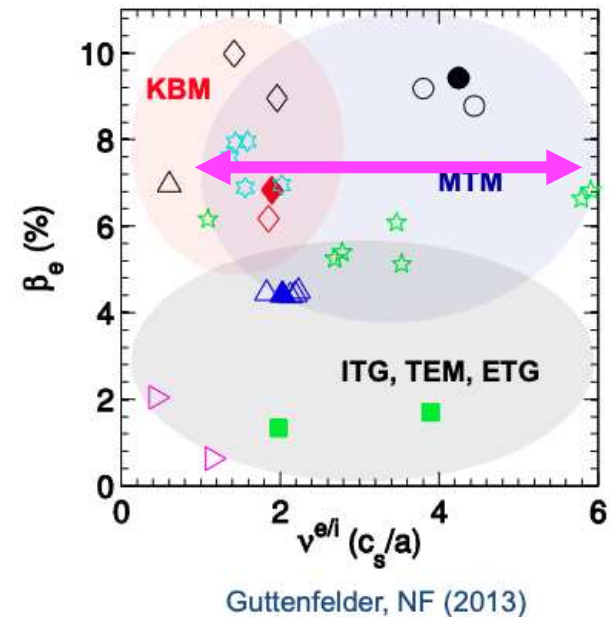
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Electromagnetic turbulent transport degrades confinement

Degradation of confinement performance of fusion devices due to electron thermal transport !

- ❑ The transport due to the electromagnetic turbulence in high beta devices, $\beta = \text{plasma press.} / \text{mag. press.}$.
- ❑ Many gyrokinetic simulations using experimental data have predicted the transport occurs by the microtearing mode (MTM)
[Wong+, 2007, Guttenfelder+ 2013, Moradi+, 2013, e.g.]



Microtearing mode

- ❑ The microtearing mode (MTM) is driven by the electron temperature gradient, accompanied by magnetic reconnection, and stable in both the collisionless and the collisional limits [Haseltine+, 1975, Drake+, 1977].
- ❑ The energy dependent electron-ion collisions leading to the thermal force is indispensable for the mode instability in a slab [Gladd+, 1980, Hassam+, 1980].
- ❑ Recent theoretical work claims the energy dependent collisions is not necessary to destabilize MTM with the reduced gyrokinetic model [Zocco+, 2015].
- ❑ Collisionless reconnecting mode is, an electromagnetic ETG mode with tearing parity [Zocco+, 2015, Geng+, 2020]. / a collisionless MTM [Kobayashi+, 2015].
- ❑ Collisionless MTM has been demonstrated in fusion devices with more realistic magnetic geometry [Dickinson+, 2013].

Questions

- ❑ The destabilization mechanism of MTM has not been fully understood even in a slab.
 - ❑ We investigate the following questions in a slab MTM by linear gyrokinetic simulations using AstroGK code [Numata+, 2010].
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- What is the destabilization mechanism of MTM ?
 - Is there collisionless MTM in a slab ?

Simulation setup

- ❑ 2-D slab model ($\partial_z = 0$).

- ❑ The equilibrium magnetic field configuration:

$$\mathbf{B}_0 = B_{z0}\hat{\mathbf{z}} + B_y^{\text{eq}}(x)\hat{\mathbf{y}}, \quad B_{z0} \gg B_y^{\text{eq}}(x).$$

- ❑ $B_y^{\text{eq}}(x)$ is Initial sheared mag. field, imposing a small perturbation with wavenumber k_y . Tearing instability parameter $\Delta'a < 0$ for MTM.

Tearing sim. setup on AstroGK
[Numata+, 2011]

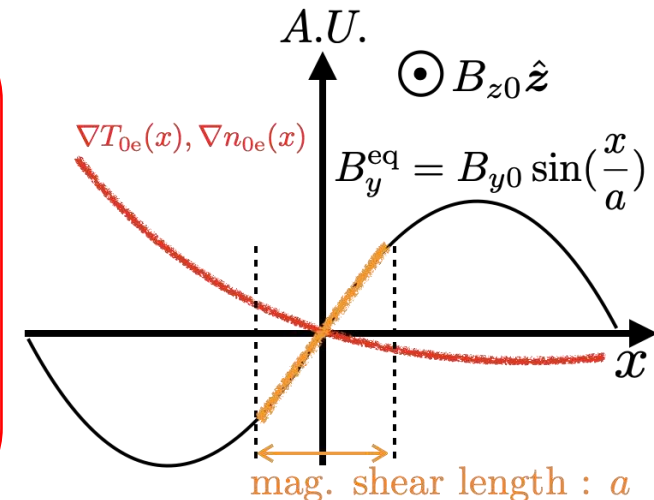
- ❑ Electron temperature and density gradients are given by

$$L_{T_{0e}}^{-1} = -\partial(\ln T_{0e})/\partial x, \quad L_{n_{0e}}^{-1} = -\partial(\ln n_{0e})/\partial x,$$

- ❑ Electron diamagnetic drift freq. is defined as

$$\omega_{*T_{0e}, n_{0e}} = k_y \rho_e v_{\text{th},e} / 2L_{T_{0e}, n_{0e}}, \quad \omega_{*e} = \omega_{*T_{0e}} + \omega_{*n_{0e}}$$

where ρ_e and $v_{\text{th},e}$ are the electron Larmor radius and thermal velocity, respectively.

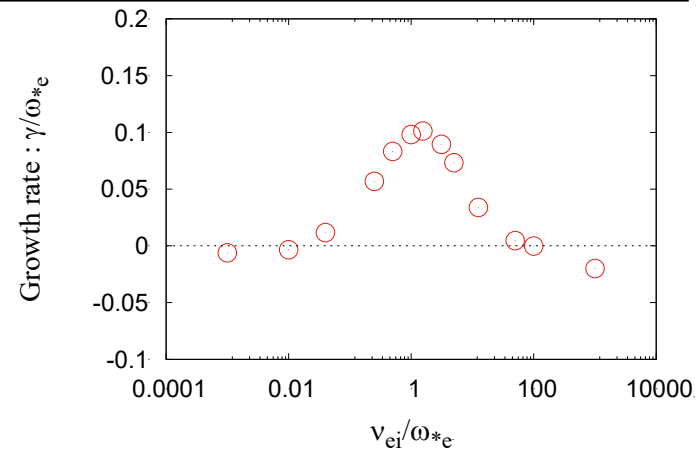
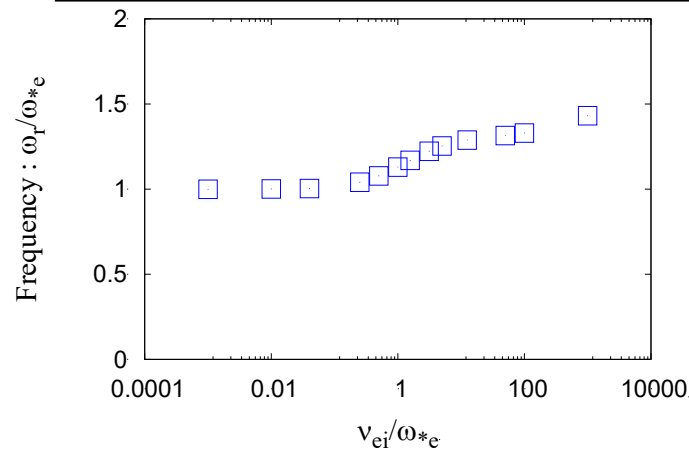


- ❑ Fokker-Planck collision operator is employed, and the energy dependent collision model is used in our simulations.

Collision frequency dependence

- ❑ The MTM is unstable in only $\nu_{ei} \sim \omega_{*e}$ regime, stable in both collisionless and collisional limits, and the mode propagates with $\omega_r \sim \omega_{*e}$.
- ❑ Collisional and drift-wave mode !
- ❑ Our results qualitatively agree with the theoretical solution [D'Ippolitto+, 1980, e.g.], however quantitative agreement has not yet been obtained.

Basic parameters : $\beta_e = 0.024$, $\Delta' a = -4.58$, $m_e/m_i = 1/1836$,
 $n_{0e}/n_{0i} = 1$, $T_{0i}/T_{0e} = 1$



Reconnection electric field by thermal force

□ We consider the electric field parallel to the magnetic field $E_{\parallel}$.

□ By the generalized Ohm's law,

$$E_{\parallel} = -ik_{\parallel}\phi + i\omega A_{\parallel} = -i\omega d_e^2 j_{\parallel} + i\omega_{*e} A_{\parallel} + ik_{\parallel} \frac{T_{0e}}{q_e} \frac{\delta P_e}{P_{0e}} + \boxed{E_{\parallel}^{\text{coll}}}$$

Ele. field due to collision.

□ [Zocco+, 2015] derived as $E_{\parallel}^{\text{coll}} = \boxed{\eta j_{\parallel}}$ by a simplified collision model.
 $= 0.38\mu_0\nu_{ei}d_e^2$: Spitzer resistivity.

□ $E_{\parallel}^{\text{coll}}$ obtained by AstroGK simulations can be written as

$$E_{\parallel}^{\text{coll}} \approx \eta j_{\parallel} + E_{\parallel}^{\text{th}} \quad (: \text{thermal force}) \quad (1)$$

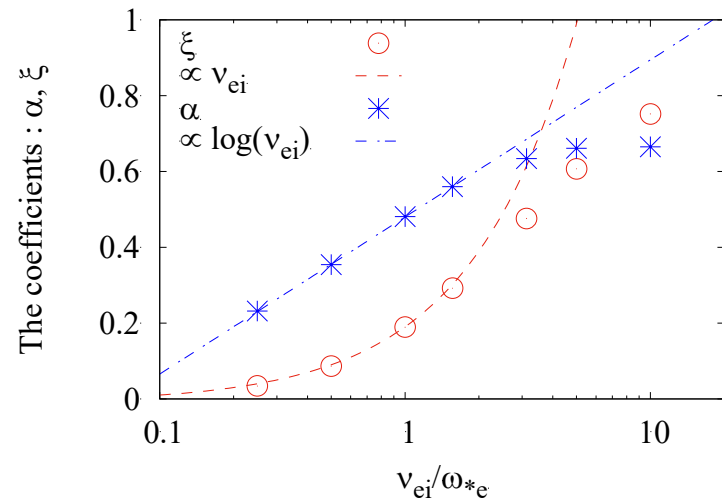
□ Under assumption $\nu_{ei} \gg \omega_{*e}$, $E_{\parallel}^{\text{th}}$ is derived by [Hassam, 1980] as follows:

$$E_{\parallel}^{\text{th}} = iA_{\parallel}\omega_{*T_{0e}}\alpha \left[-\frac{\omega}{\nu_{ei}} A_{\parallel}\omega_{*T_{0e}}\xi \right], \quad (\alpha \text{ and } \xi : \text{coefficients.}) \quad (2)$$

Time dependent thermal force term : $E_{\parallel}^{\text{ttf}}$

$E_{\parallel}^{\text{ttf}}$ is believed to play an important role in MTM, not the first term of $E_{\parallel}^{\text{th}}$.

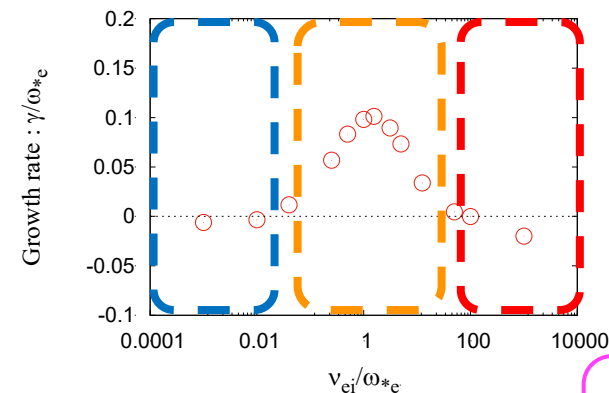
Evaluation of thermal force on MTM



$$E_{\parallel}^{\text{th}} = iA_{\parallel}\omega_{*T_{0e}}\alpha - \frac{\omega}{\nu_{ei}}A_{\parallel}\omega_{*T_{0e}}\xi, \quad \nu_{ei} \gg \omega_{*e} \quad (2)$$

$= E_{\parallel}^{\text{ttf}}$

- α and ξ are measured from our simulation.
- α and ξ are not constants in $\nu_{ei} \lesssim \omega_{*e}$, become constants in $\nu_{ei} \gg \omega_{*e}$ regime.



In $\nu_{ei} \rightarrow \infty, \alpha, \xi = \text{const}$, hence, $E_{\parallel}^{\text{ttf}} \rightarrow 0$: MTM stable.

In $\nu_{ei} \rightarrow 0, \xi \rightarrow 0$, hence, $E_{\parallel}^{\text{ttf}} \rightarrow 0$: MTM stable.

Only in $\nu_{ei} \sim \omega_{*e}$ regime, $E_{\parallel}^{\text{ttf}} = \text{finite}$: MTM unstable.

The destabilization of MTM with respect to ν_{ei} is consistently described by time-dependent thermal force !

Summary

- ❑ We have performed linear gyrokinetic simulations of a 2-D slab MTM using AstroGK.
- ❑ The MTM is collisional and drift-wave mode, and unstable only in $\nu_{ei} \sim \omega_{*e}$ regime.
- ❑ The time-dependent thermal force is indispensable to destabilize the mode, namely, the reconnection electric field is supported by the thermal force !
- ❑ Collisionless MTM cannot exist in terms of the thermal force.
- ❑ Why does the time-dependent thermal force contribute to the mode destabilization?